

Networking Basics

IP Address helps us uniquely identify different machines over the pub network

IP Address:-
IPv4 - 32 bit (4 billion)
012.345.678.910 client
255.255.255.255 (max)
IPv6 - 128 bit (max number of IP Address)



Server (Resembles packets)

(Say want to send the pages of the book)

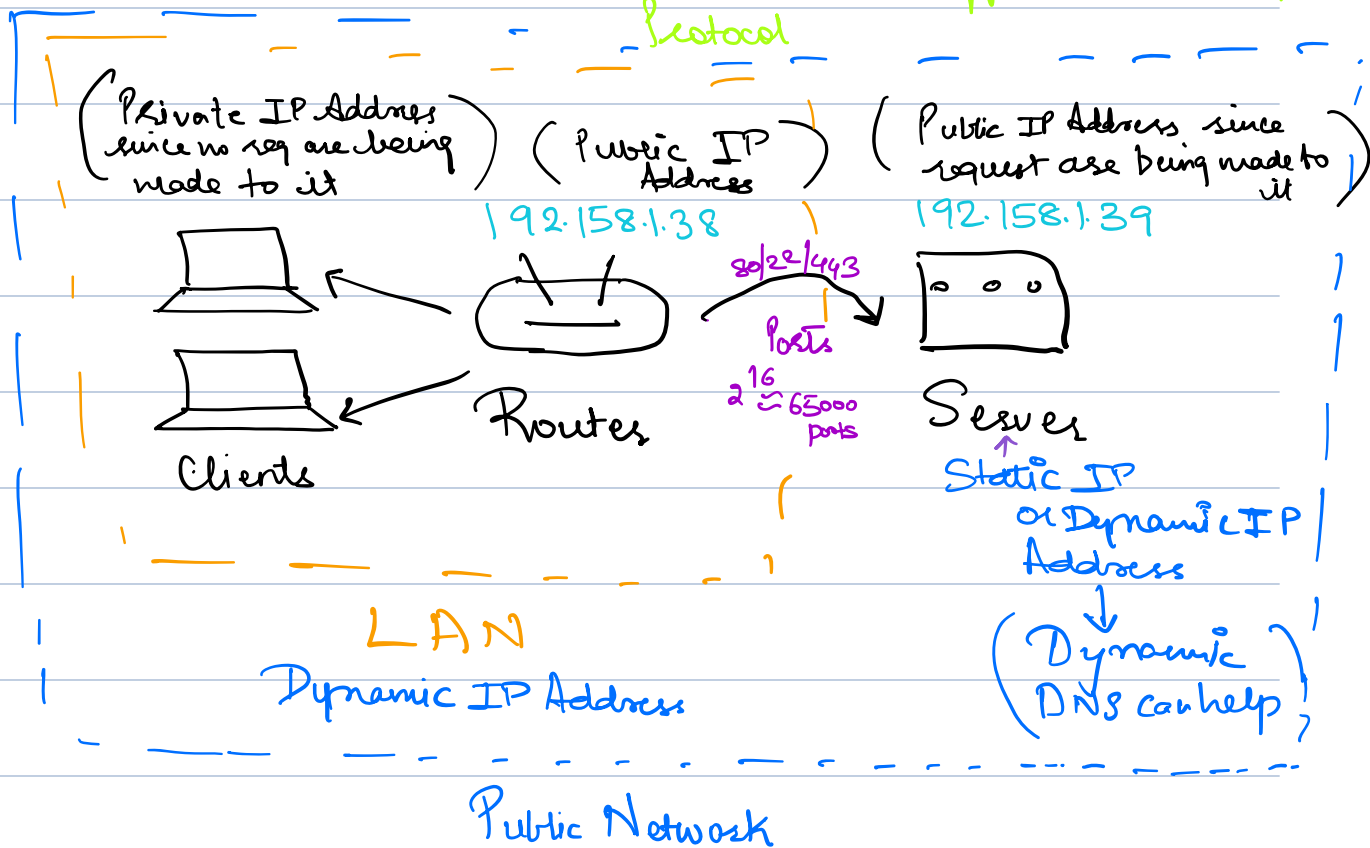
% To understand the the exact purpose of TCP

IP Protocol:- Is concerned with the network layer

watch the video at 05:41

TCP Protocol:- Is concerned with the transport layer (To Resemble packets)

HTTP Protocol:- Is known as Application layer Protocol



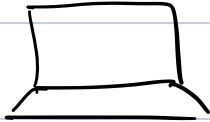
Local Host $\rightarrow$ 127.0.0.1 Reserved IP Address accessing the server that is running locally on your own machine

Ports ^{channel for communication} Is where my application is running (16 bit value $\approx 2^{16} = 65536$) Also, single application uses single port

Eg:- HTTP (default) - 80 port, HTTPS - 443 port (used implicitly)

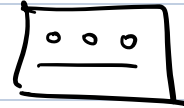
TCP and UDP

192.158.1.38



Client

192.158.1.39



Server

Internet Protocol
suite (TCP/IP)

TCP (Transmission Control Protocol) :- Helper in handling the order of the packets (Reliability - Retransmission of the lost packets)
(HTTP, SMTP, WS) (Web Response) (Mails)
Establishes a connection (3 way Handshake) so data can be sent from both the sides

Disadv :- Latency

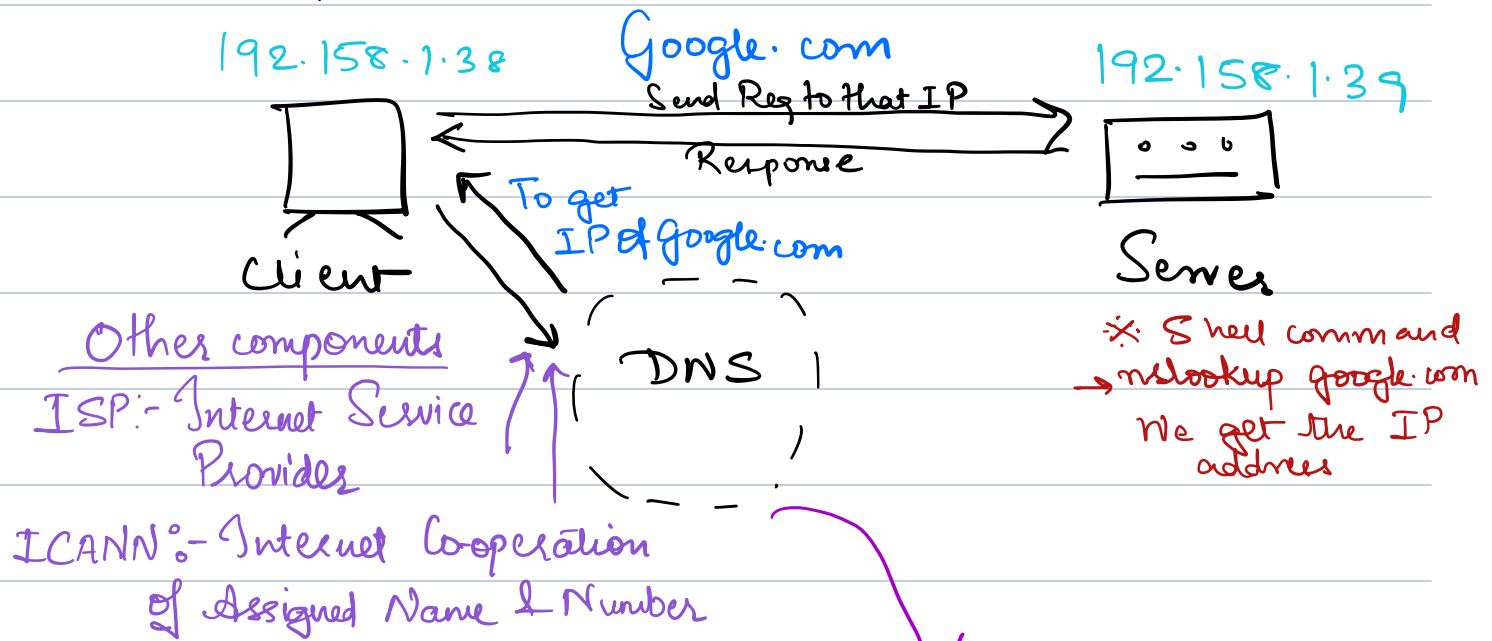
Overhead component makes it expensive because to have 2-way connection we need to first establish 3-way handshake

UDP (User Datagram Protocol) :- No need to establish a connection between client & server (eg not 3-way handshake needed)
DNS, live streaming, Online Games

Just Faster

Opposite to TCP For Eg:- like Req is sent from client and server just starts sending data so any data packets missing won't be resend again to client
→ No ordering (Of course some ordering will be there)

DNS Domain Name System



% Client will go through ISP (Internet Service Provider) then will go to ICANN which will go into the Registrar to check the corresponding domain server (Google domain) that server will hold the DNS Record

DNS Record.

Address → IP

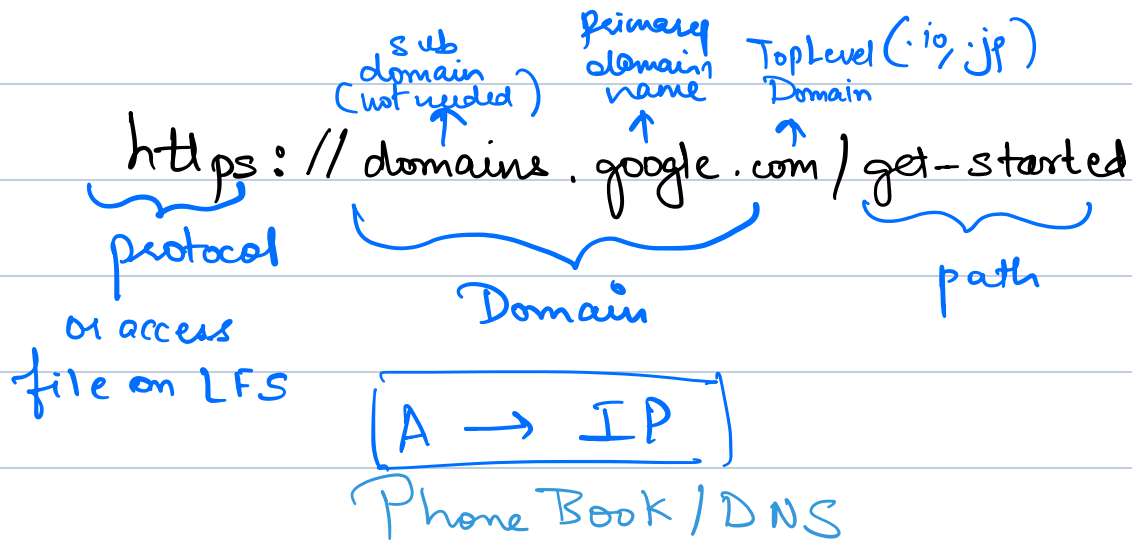
ICANN - Organization that owns all the domains

Domain Registrar - Resellers of those domains host those servers that have DNS Records (Google domain, Godaddy)

Caching - Can help avoid going to DNS as the after one request we can have IP Address

Saved on our System

Firewall - configuration helps to either avoid getting request on public service or either allow it



Local Example for Better Understanding

So the AT&T wifi router is where when my device wants to connect to internet it will help first connect to the router through the password which in turn will help to to connect to the internet cable/internet

Yes, you have the right idea. Let me simplify the flow:



Step by step:

- 1. Your device connects to the AT&T Wi-Fi router**
 - You select your Wi-Fi name (SSID).
 - You enter the Wi-Fi password.
 - Your device is now connected to your home network.
- 2. The router connects your home network to the internet**
 - The router is already connected to AT&T's internet line (fiber/cable).
 - It sends your requests out to the internet.
- 3. The response comes back**
 - Website server → AT&T network → your router → your device.

Example:

your **laptop, phone, and iPad** are connected
to the same Wi-Fi router, they are part of the
same **local network (LAN — Local Area
Network)**.

home router gives each device a private IP address inside your local network.